# XO-1b

**R-0091**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_XO-1_180524` · created 2026-07-15T22:33:57+00:00

**BENCHMARK: INACCURATE**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2458262.8251 +/- 0.0035 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.11 +/- 0.032 |
| Transit depth (Rp/Rs)^2 | 1.21 +/- 0.7 [%] |
| Orbital Inclination (inc) | 89.94 +/- 1.16 |
| Airmass coefficient 1 (a1) | 0.9881 +/- 0.0057 |
| Airmass coefficient 2 (a2) | 0.009 +/- 0.021 |
| Scatter in the residuals of the lightcurve fit is | 0.58 % |
| Best Comparison Star | #2 - [257, 179] |
| Optimal Aperture | 5.3 |
| Optimal Annulus | 13.26 |
| Transit Duration (day) | 0.0203 +/- 0.0046 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.11 ± 0.032 vs 0.1326 (deviation 0.71σ) |
| Duration fit vs published | 0.0203 d vs 0.1238 d (deviation 22.49σ) |
| Mid-time O–C | 9.8 min |
| Depth SNR | 3.4 |
| Residual scatter | 0.58 % |

## Light curve

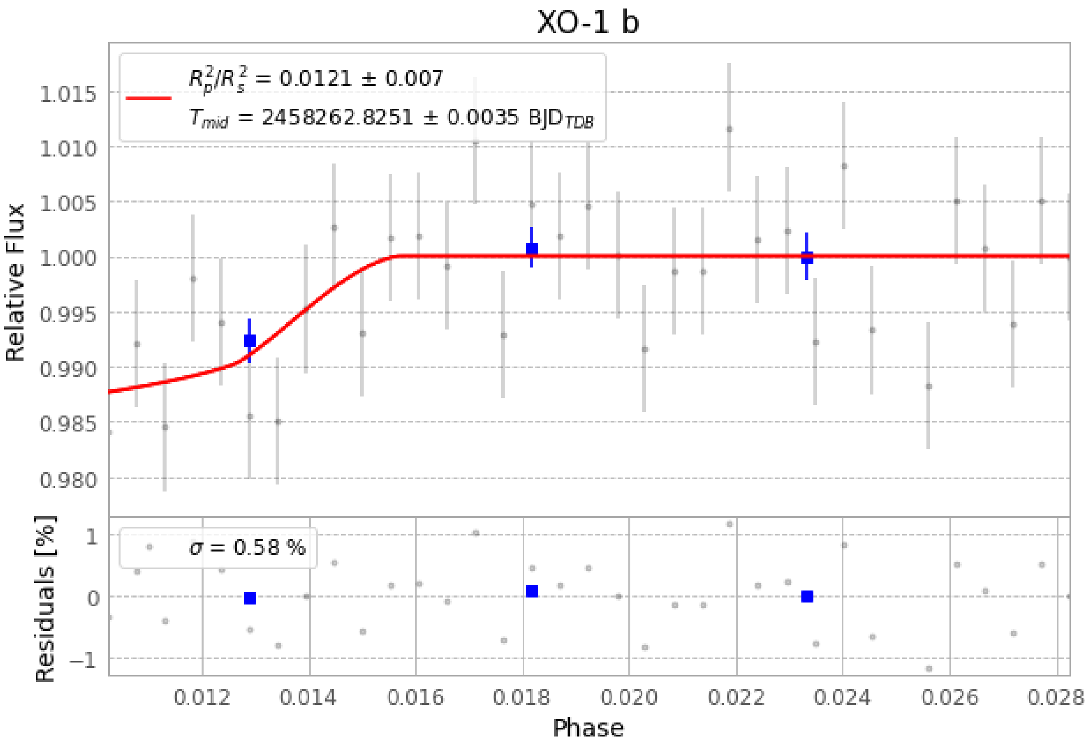


Figure 1 — FinalLightCurve_XO-1 b_2018-05-24.png (SHA-256 493cd520bb4d3ea1…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [246, 281], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 54bb46d2b4a725d9…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

## Data provenance

35 FITS frames (23 MB), `XO-1180524083912.FITS` → `XO-1180524102112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-180524.log (4b67888d7a21…), FinalLightCurve_XO-1 b_2018-05-24.png (493cd520bb4d…), FinalLightCurve_XO-1 b_2018-05-24.csv (ddb5929a02d8…)

## Compute resources

Wall time 100.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)